Chem 150 **Answer Key** **Term Test 1 A**

1.(**2 marks**) Using the periodic table arrange the following elements in the expected order of increasing size: Ca, Se, Br, Sr

Br < Se < Ca < Sr

2.(**3 marks**) Arrange the following ions in order increasing size: **S^{2-}, K^{+}, Se^{2-}, Ti^{4+}**

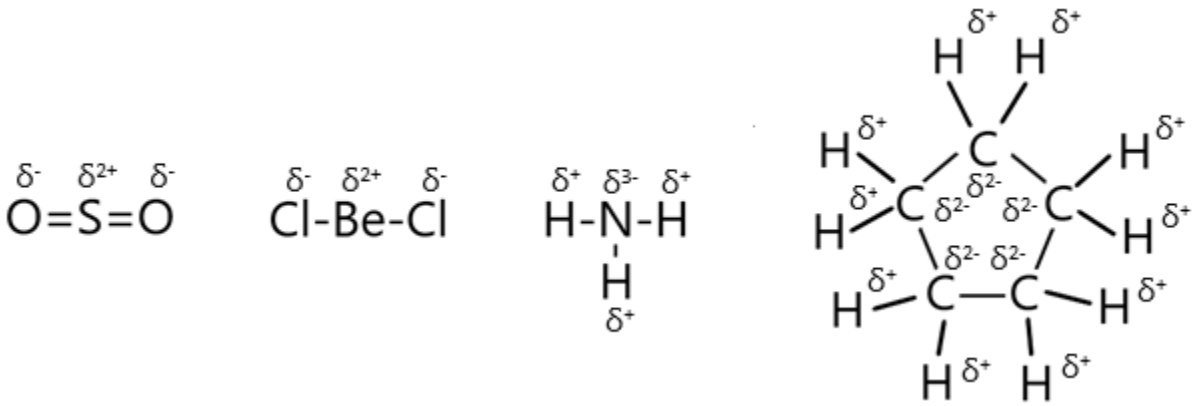
species	**S^{2-}**	**K^{+}**	**Se^{2-}**	**Ti^{4+}**
# of protons	16	19	34	22
# of electrons	**18**	**18**	36	**18**

Arrange isoelectronic ions first! The greater the number of protons, the smaller the ion. **$Ti^{4+} < K^{+} < S^{2-}$**

Next look at the "oddball" **Se^{2-}** which is in the same group as **S^{2-}** but has a higher atomic number meaning there are more electronic shells filled in **Se^{2-}** compared to **S^{2-}** so **Se^{2-}** will be bigger than **S^{2-}**.

Final answer: **$Ti^{4+} < K^{+} < S^{2-} < Se^{2-}$**

3. (**6 marks**) Consider the molecules below. Using the electronegativity values provided on the last page of this exam, assign partial (fractional) charges (δ^{+}, δ^{-}, δ^{2-} etc.) to ALL atoms in the four molecules.



4.(**4 marks**) Write balanced equations for the reactions of a) sulfur with aluminum and b) magnesium with chlorine.

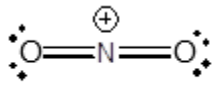
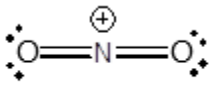
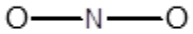
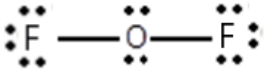
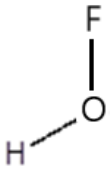
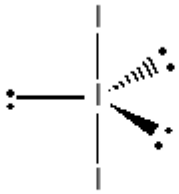
a) _____$2Al + 3S \rightarrow Al_2S_3$_____

b) _____$Mg + Cl_2 \rightarrow MgCl_2$_____

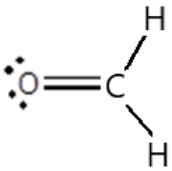
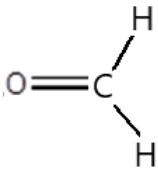
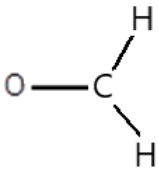
5. (**10 marks**) Draw four valid Lewis structures of the phosphate ion $[PO_4]^{3-}$. If there are formal charges on any of the atoms be sure to include them! Indicate if the structures you have drawn are equivalent or non-equivalent. Circle equivalent Resonance structures. If you have non-equivalent resonance structures, state this also and underline your best resonance structure.

See answer Key Problem set 2

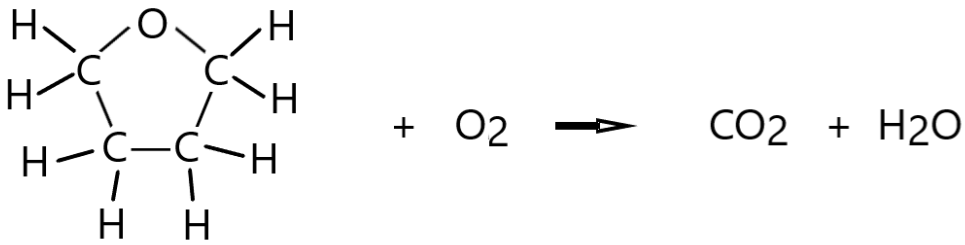
4.(**21 marks**) Identify the electron group geometry (EGG) and the molecular shape of the species below. Start by writing a valid Lewis structure and state the number of electron groups (#EG) around the central atom. Be sure to show all electrons and indicate any non-zero formal charges. Continue by drawing the EGG placing lone pairs and bonding pairs in the appropriate positions. **Name the EGG**. Next draw the molecular shape. Here multiple bonds, formal charges on atoms and lone pairs might be omitted for clarity. **Name the molecular shape**. Finally, indicate if the molecule or ion is polar or not. Please refer to the example below (SF_4) if in doubt about any of the instructions.

Lewis Structure	#EG	EGG	shape	polar (Yes or No)
	2	 linear	 linear	No
	4	tetrahedral	 bent	Yes
	5	 trigonal bipyramidal	linear	No

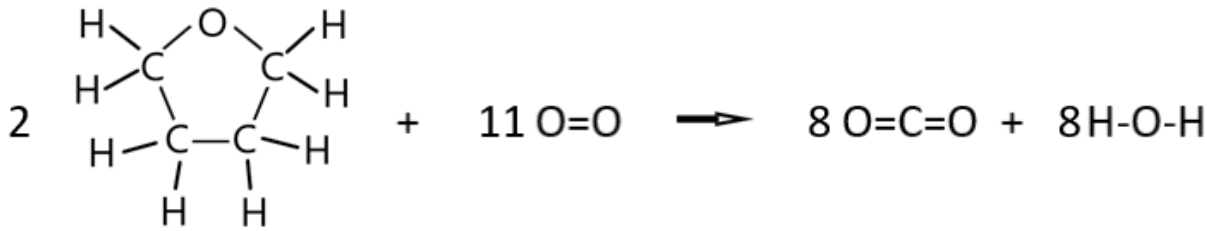
Hint: Only use single bonds between I and I_____

	3	 trigonal planar	 trigonal planar	Yes

6.(**6 marks**) The molecule shown below is tetrahydrofuran (THF), a cyclic organic ether which is extremely flammable and combusts readily with oxygen to form carbon dioxide and water.



a) Balance the equation above: **$2C_4H_8O + 11O_2 \rightarrow 8CO_2 + 8H_2O$**



b) Using bond dissociation values provided in the table below estimate the energy change that occurs when **one** mole of THF combusts quantitatively. Hint. You might want to draw Lewis structures off all reactants and products.

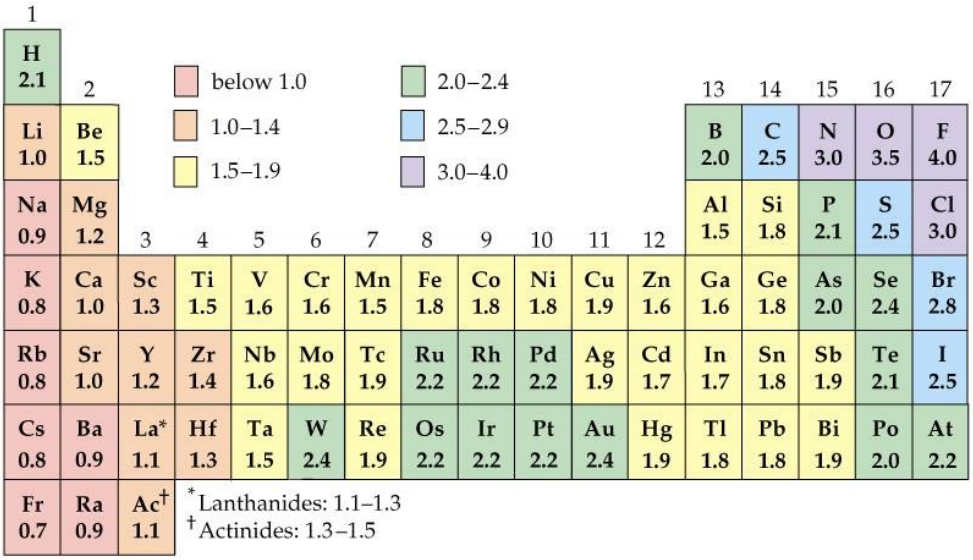
Energy required to break bonds:	**Energy released forming bonds**
2x2mol x 360 kJ/mol for 4 C-O bonds = 1440 kJ	8x2mol x 799 kJ/mol for 16 C=O bonds = 12784 kJ
2x3mol x 347 kJ/mol for 6 C-C bonds = 2082 kJ	8x2 mol x 464 kJ/mol for 16 H-O bonds = 7424 kJ
2x8mol x 414 kJ/mol for 16 C-H bonds = 6624 kJ	
11mol x 498 kJ/mol for 11 O=O bonds = 5478 kJ	
15624k J	20208 kJ

More energy is released than consumed!

20208 kJ – 15624 kJ = 4584 kJ are released for the combustion of two mole of THF so half would be released per one mole. **Final answer: 2292 kJ/mol are released.**

c) State if energy is absorbed by the reaction or if it released (produced).

Energy is released (produced) by the reaction.

bond	d_0 (kJ/mol)	bond	d_0 (kJ/mol)	bond	d_0(kJ/mol)
H-H	436	C-C	347	N-N	163
H-C	414	C=C	611	N=N	418
H-N	389	C≡C	837	N≡N	946
H-O	464	C-N	305	N-O	222
H-S	368	C=N	615	N=O	590
H-F	565	C≡N	891	O-O	142
H-Cl	431	C-O	360	O=O	498
H-Br	364	C=O	799	Cl-Cl	243

Electronegativities of the elements